Predicting Flight Delays Across U.S. Airports

CSE 163 Final Project Report

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Public repository:

https://github.com/yalharbii/flight-delay-eda

This report analyzes 10,000 U.S. flight-delay records and evaluates airport, route, scheduling, distance, and prediction patterns.

1. Introduction

Delays are a fairly typical occurrence for flights, but the extent and distribution of such events is not equal for all flights. A flight delay may occur due to some interaction between processes occurring at an airport, route choice, flight scheduling, passenger demand, and other processes which are not apparent to the customer. In order to detect the patterns in which the delays usually concentrate and also to measure the degree of difficulty of predicting a very long delay based on basic flight data.

In the current research, a dataset of 10,000 flights will be analyzed in order to explore delay patterns related to airport, route, day of the week, hour, and flight distance. In addition, using machine learning techniques, a random-forest classifier will be created in order to see if flight origin, destination, scheduling data, and distance will be able to predict a 30-minute or longer delay. This research does not seek to create an operational forecast tool, rather an attempt to integrate descriptive and machine learning analysis in one workflow.

2. Summary of Questions and Results

1. Which airport, time-of-day, day-of-week, and distance patterns are most associated with longer flight delays? However, the most pronounced patterns found were related to the time and the day of the flight that was planned. The highest average delay was observed for Friday (16.55 minutes), and 20.23% of all flights that took place on Fridays faced a delay of over 30 minutes. Moreover, late-night flights experienced significantly higher average delays (24.38 minutes) as opposed to morning flights (0.96 minutes). Shorter flights within the sample were characterized by higher average delays compared with longer ones.

2. Which origin airports, destination airports, and frequently observed routes had the largest average delays? Among the destinations with more than 20 observations, HRL had the highest average delay (24.38 minutes), followed by CRP and BOI. Origins with the highest average delays include SFO (15.79 minutes) and SJC (15.48 minutes). Finally, HOU-HRL had the highest average delay rate among all routes (29.00 minutes) based on 40 flights in total.

3. How well can basic flight characteristics predict whether a flight will be delayed by more than 30 minutes? Random Forest model resulted in 61.5% of accuracy, 19.5% of precision, 74.8% of recall, and F1 score of 0.309. Despite the relatively high level of recall (almost 75% of all cases that experience delay of over 30 minutes are detected by the model), the low level of precision suggests that it should be treated rather as an exploratory model.

3. Motivation

Flight delays impact both the passengers and the airlines, but what happens during a delay seems to be a matter of chance to a traveler. The field of aviation interests me and I thought it would be a good chance to analyze a real-life system, which also has significant practical implications. In contrast to the traditional approach of viewing the delays as the national average, the current project tries to determine if time of departure, airports, routes, and distances impact the probability of a delay differently for various flights.

In addition, this project allows distinguishing between describing and predicting using a particular system. While the group summary will show which flights had higher delays in this dataset, a classification model can help to test the hypothesis of using available flight parameters to predict the probability of delay more than 30 minutes. The distinction is important since an interesting descriptive finding does not necessarily mean the possibility of constructing a predicting model.

4. Dataset

The final analysis utilizes the `flights_10k` dataset of the Vega Datasets. According to Vega, this resource comprises 10,000 flight delay records created based on the U.S. Bureau of Transportation Statistics (BTS) Reporting Carrier On-Time Performance database. The downloaded file includes five original attributes: `date`, `delay`, `distance`, `origin`, and `destination`. The records in the downloaded sample cover the period from January 1 through March 31, 2001.

Variable	**Meaning**
date	Flight date and time associated with the record
delay	Flight delay in minutes. negative values indicate an early flight
distance	Flight distance in miles
origin	Origin airport code
destination	Destination airport code

The dataset comprises 10,000 observations, 59 distinct origin airport codes, 59 distinct destination airport codes, and 643 distinct origin-destination pairs. All of the five original attributes are filled, no missing values have been identified. The mean flight delay equals 8.11 minutes; the median is 0 minutes. The minimum and maximum delays are -55 and 377 minutes respectively. The mean flight distance is 501.34 miles; the minimum and maximum distances are 108 and 2,298 miles respectively.

Totally, 1,150 flights, or 11.5% of the whole sample size, have delays more than 30 minutes. This imbalance is important for the task of predictions since a naive model predicting "not more than 30 minutes" for all the flights would achieve 88.5% accuracy. In this context, the accuracy metric has to be interpreted along with the precision, recall, and F1 scores.

5. Method

The codebase is separated between two primary Python files: `eda.py` includes code for loading the data, feature engineering, summary statistics, and exploratory visualization; `analysis.py` includes comparison of days of the week and routes and the prediction model. `download_data.py` ensures reproducibility of the data loading process by downloading Vega JSON file and saving the exact same data to `flight_sample.csv`.

There were five engineered features used in the analysis. `delay_over_30` is a Boolean variable taking True value if `delay` was greater than 30 minutes. The datetime column is parsed as pandas datetime object and used to create `hour`, `day_of_week` and `month` features. The scheduled hours were categorized as: Overnight, Morning, Afternoon, Evening and Late night.

Airports are compared excluding those with less than 20 observations, and the same cut-off value is used for route comparison. These filters decrease chances of an airport/route being extreme due to a small number of observations, but do not account for sampling error. There are four distance categories used in the analysis: 0-500, 501-1000, 1001-1500, 1501-2500 miles.

In the prediction analysis, categorical variables `origin`, `destination`, `time_of_day` and `day_of_week` were converted to indicator columns via `pandas.get_dummies` function. In addition, `month`, `hour` and `distance` were included in the model. The data set was partitioned into 80/20 train/test subsets where the proportion of 30+ minute delays was the same in both subsets, stratification was done. Random-forest classifier was trained with 100 trees, maximum tree depth equal to 10, `class_weight="balanced"` and `random_state=1`. Balanced class weights were used since long delays were in minority.

6. Exploratory Data Analysis

In the EDA process, it was confirmed that there are no missing values out of the 10,000 observations. The summary statistics revealed that the delay distribution is extremely right skewed since the median equals zero, the mean equals 8.11 and the maximum equals 377. This indicates that a few very large values pull the mean value. It should also be noted that the values include negative delays, with the minimum delay being -55 minutes.

The sample contains relatively short flights, with 6,840 flights being less than or equal to 500 miles and 272 flights between 1,501 and 2,500 miles. Such information influenced the decision to provide group sizes along with the mean delays as well as not to interpret the relationship between distances and delays as causality.

There is also great variation in terms of scheduled times since the flights in the morning time period are the most numerous and have the lowest mean delays among other categories of the time of scheduled flight.

7. Results

Research Question 1: Timing and Scheduling Patterns

There is considerable variance related to day of the week in this sample. Figure 1 reveals that the maximum mean delay occurs on Fridays, which is 16.55 minutes, followed by Thursdays, which have a mean delay of 9.61 minutes. On Mondays, there is a minimum mean delay of 4.11 minutes. The maximum percentage of delays exceeding 30 minutes occurred on Fridays at 20.23%.

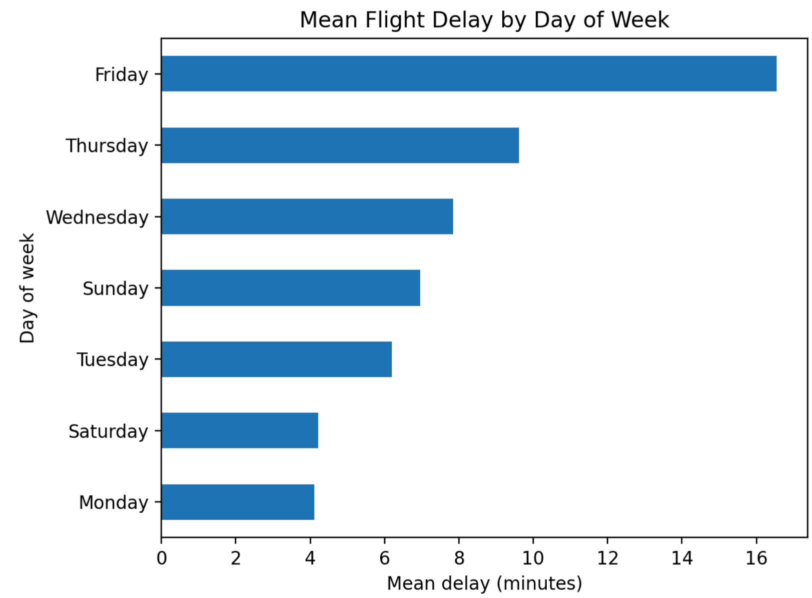

Figure 1. Mean flight delay by day of week. The plot is generated by analysis.py from all 10,000 records.

Time of day made an even bigger difference. Morning flights were delayed on average by only 0.96 minutes with only 3.68% of them delayed for more than 30 minutes. This number increased in the afternoon flights to 9.61 minutes, and for evening flights it was 12.30 minutes. Late night flights were delayed on average by 24.38 minutes, and 29.94% of them were delayed for more than 30 minutes.

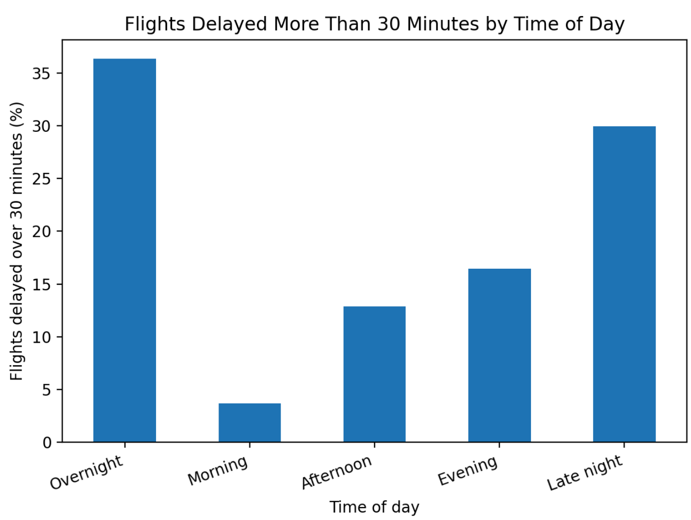

Figure 2. Share of flights delayed more than 30 minutes by scheduled time of day, generated by eda.py.

Research Question 1: Distance Pattern

In the case of the delay versus distance comparison, an increase in distance resulted in a decrease in the average delay time. While 0-500 miles resulted in an average delay of 9.18 minutes, 501-1,000 miles had an average delay of 6.96 minutes, 1,001-1,500 miles had an average delay of 4.50 minutes, and 1,501-2,500 miles had an average delay of 0.18 minutes

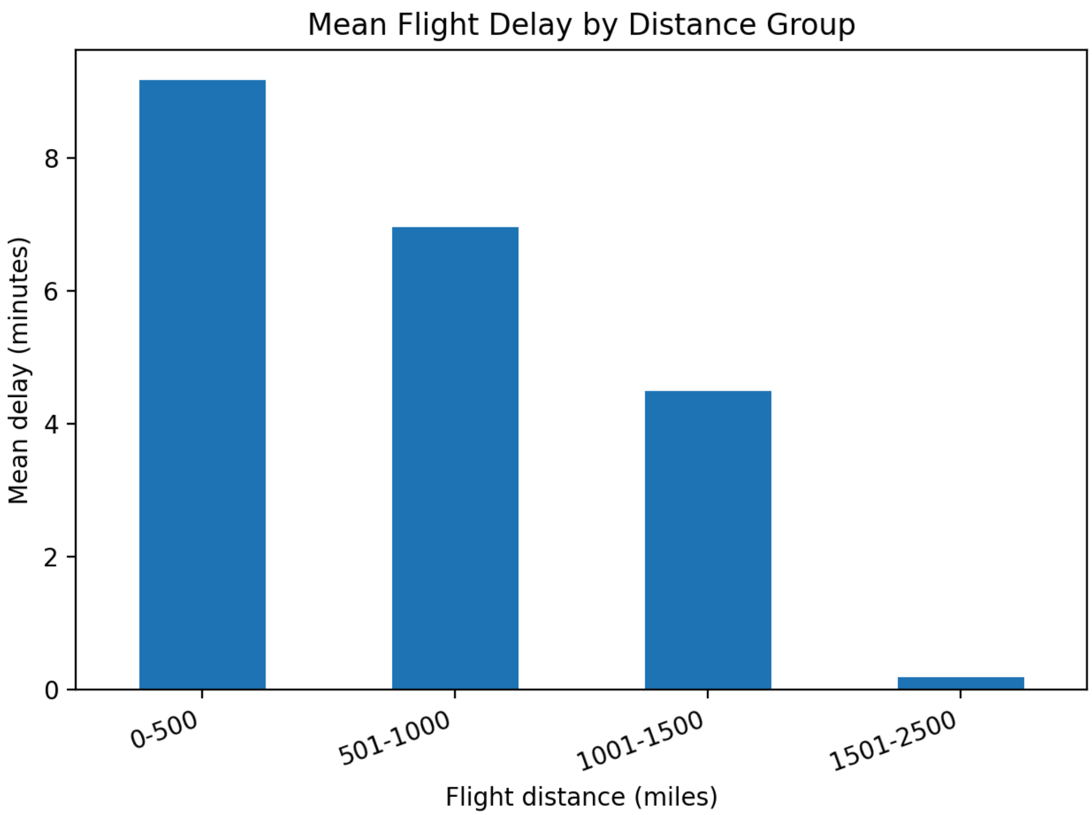

Figure 3. Mean flight delay by distance group. The chart is generated by eda.py using four distance bins.

The above findings should not be taken as proof that longer distances between departure and arrival points help avoid delays. Distance has something to do with the routes taken, airports used, timing of flights, and types of flights in the sample. The longest-distance sample has only 272 cases, while there are 6,840 short-distance flights in the data set.

To answer Research Question 1, the above findings suggest that the best patterns found in the descriptive analysis refer to time of flights and certain days. Fridays, late-night departures, and the smaller overnight sample are the clearest in the analysis, along with the distance pattern whose interpretation is less clear due to differences in size and route mix.

Research Question 2: Airports

Summary statistics were provided for airports with a minimum of 20 data points. For destination airports, the highest average delay was found at HRL with an average delay of 24.38 minutes based on 52 flights. CRP was second with an average delay of 17.10 minutes based on 30 flights, followed by BOI with 15.91 minutes based on 70 flights. HRL also had a high 30+ minute delay rate of 30.77%.

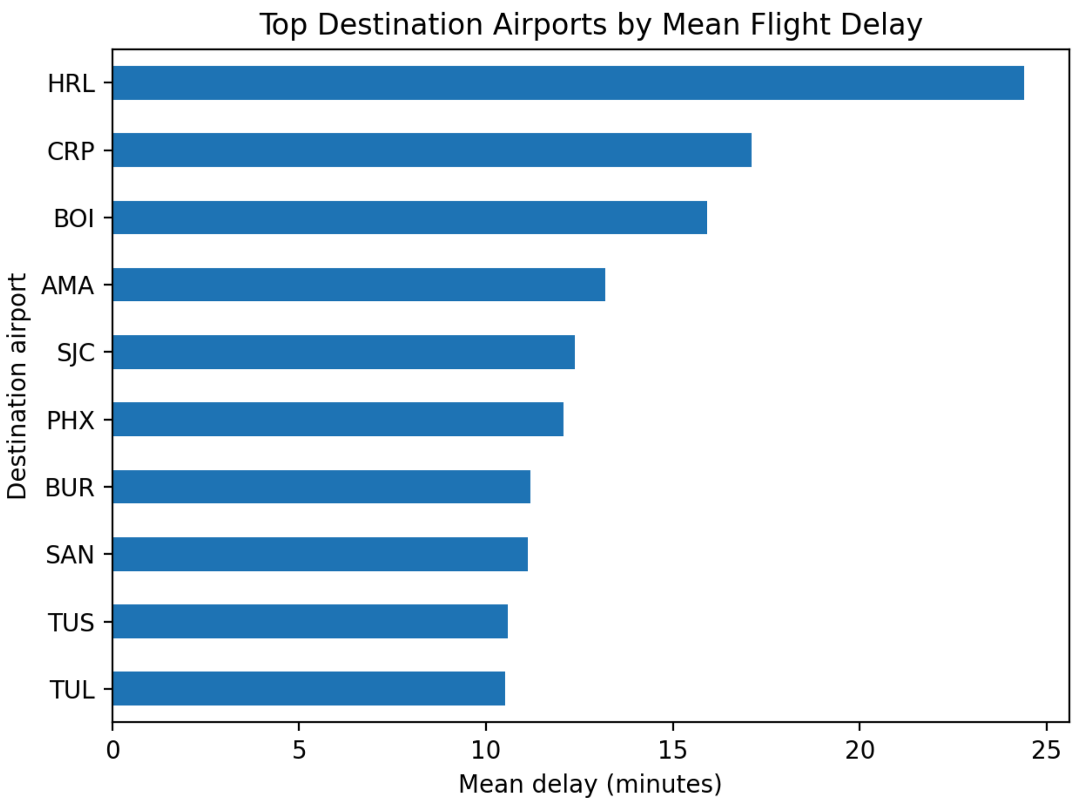

Figure 4. Top destination airports by mean flight delay among airports meeting the 20-flight threshold, generated by eda.py.

Origin airports' rankings were slightly different. The mean origin delays of the top three origin airports that qualify to rank were SFO with 15.79 minutes but with only 28 samples; SJC with 15.48 minutes with 267 samples; and BHM with 14.27 minutes and 86 samples. The rankings cannot be considered as representative rankings in the whole nation because the samples from the airports are imbalanced.

Origin	**Flights**	**Mean delay**	**30+ min rate**
SFO	28	15.79	21.43%
SJC	267	15.48	17.60%
BHM	86	14.27	18.60%
IND	65	13.22	13.85%
LAX	421	12.47	17.81%

Research Question 2: Routes

Route analysis combines the information on origin and destination to create a single route code and discards any routes that have fewer than 20 observations. The following figure (Figure 5) displays the top ten routes having the highest mean delay under this condition. HOU-HRL comes first with a mean delay of 29.00 minutes for 40 flights, where 37.5% of the observations have delays more than 30 minutes.

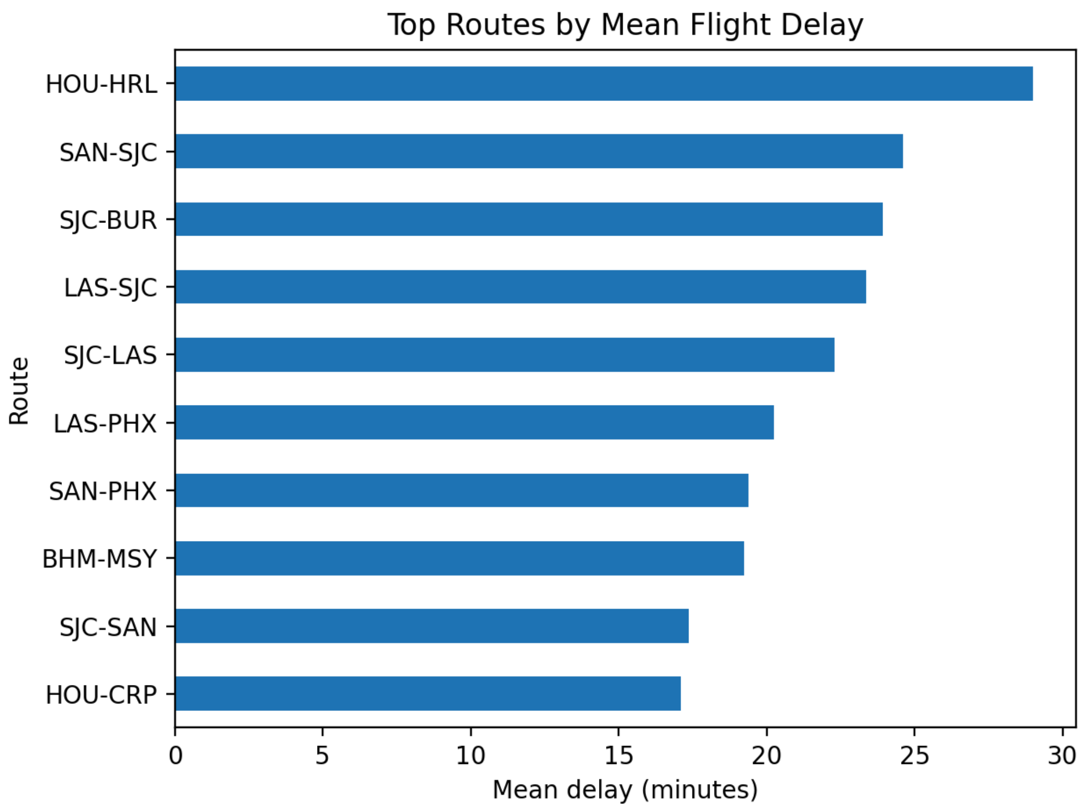

Figure 5. Ten routes with the highest mean delay among routes with at least 20 observations, generated by analysis.py.

The findings of Route Delay Analysis further emphasize the importance of the group size as well. There are routes with identical means but very different percentages of long delays. The SJC-BUR route has a mean delay time of 23.92 minutes but only 12.0% of the 25 delays go above 30 minutes, compared to the LAS-SJC route which has 32.0% of 25 flights with delay times exceeding 30 minutes despite having an almost identical mean of 23.36 minutes.

In conclusion, Research Question 2 confirms the variability of delay times among airports and routes included in the dataset. However, generalizations cannot be made without more information as the dataset covers only three months of observations.

Research Question 3: Predicting 30+ Minute Delays

The Random Forest model relies on variables available from the small data set prior to evaluating the actual result of whether there was a delay or not: origin airport, destination airport, schedule time category, day of the week, month, hour, and distance. Twenty percent of the data is included in the test set, and the proportions remain the same for each class. Since only 11.5% of the flights experience delays for more than 30 minutes, balanced class weights have been applied.

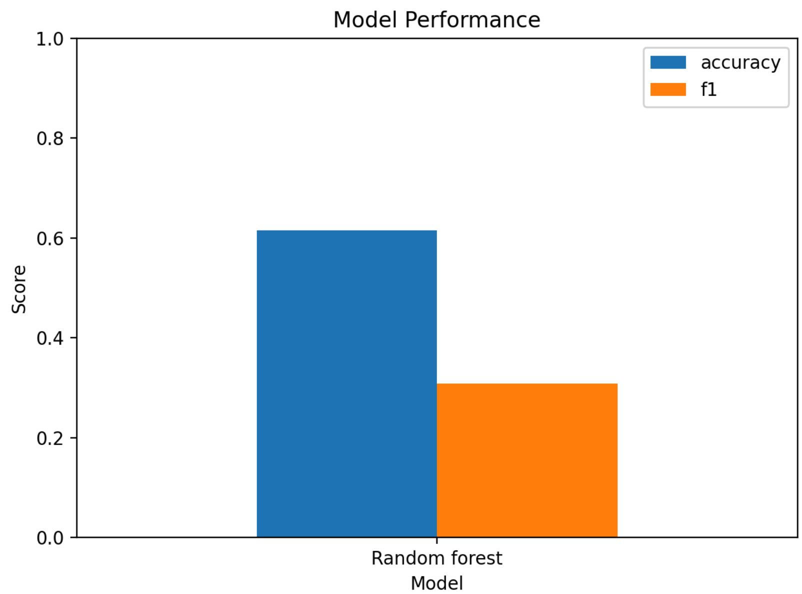

Figure 6. Random-forest test performance. The figure is generated by analysis.py; accuracy and F1 are plotted on a 0-1 scale.

Metric	Test value	Interpretation
Accuracy	0.615	61.5% of all test predictions were correct.
Precision	0.195	About 19.5% of predicted long delays were actually long delays.
Recall	0.748	The model identified about 74.8% of actual 30+ minute delays.
F1	0.309	The combined precision-recall score remained low.

Hence, the model has more emphasis on sensitivity than on precision. The model detects about three quarters of all the long delays, but the positive detections include mostly false detections. This is clearer than just the accuracy figure. There exists a classifier which can predict the majority class with an accuracy of 88.5%, hence the random forest does not perform well compared to the classifier. The value in the random forest is that the balancing weights force the model to detect long delays since it cannot ignore the minority class.

As regards Research Question 3, it is fair to say that there exists some signal in the variables, but they are not sufficient to develop a good model. Better performance would likely require variables such as airline, incoming-aircraft status, weather, congestion, scheduled versus actual operations, and a much broader training period.

8. Impact and Limitations

The most straightforward value of this study lies in education and exploration. Travelers and airport designers could use similar summaries to see when and where delays occurred in the past. Moreover, the modeling experiment demonstrates an essential principle: a highly accurate model may be unhelpful for predicting an unusual phenomenon. The report of recall, precision, and F1 gives a more comprehensive view of whether the model is detecting the target event at all.

A few limitations strongly limit the usage of these results. First of all, the final dataset contains just 10,000 transformed rows of the full BTS database. In addition, it contains just three months, which do not correspond to any recent airline networks, airport activities, and weather conditions.

Secondly, the five initial variables do not contain information about the airline, weather, reason of cancellation, aircraft rotation, runways, and air traffic control restrictions. Thus, no causation may be established, and the absence of these factors leads to the wrong predictions.

Thirdly, there are a lot of differences in the number of groups. There are just 22 observations in the overnight time category, while some of the qualifying airports contain less than 30 flights per day and even route analysis includes some small samples of 20 observations. The minimum flight thresholds may reduce the most extreme cases but cannot eliminate uncertainty. Therefore, the figures show the pattern of this sample rather than stable rankings of airports and routes.

Fourthly, the result on the distance is very susceptible to the confounding bias. Short routes occur in the dataset more often and may be associated with other factors, airports, flights schedule, and other operational patterns. It would be unreasonable to claim that distance is causing low rates of delays. In addition, the day and time of the week may influence the delays, but not cause them.

Finally, the prediction system with 19.5% precision will make many false warnings. If such a model will be offered to the travelers, it may produce the additional stress or bad travel decisions. Such deployment requires the use of new data, better features, probabilities, checking the fairness and performance of the model for each airport and carrier, and explanation of uncertainties.

9. Challenge Goals

There were initially two challenge directions proposed: the integration of multiple datasets, including external weather data, and the application of machine learning predictions. Both of these were later trimmed down, although in slightly different ways. The problem of integrating multiple datasets was trimmed due to several issues observed during the EDA phase: it turned out to be more difficult than anticipated to match weather observations to the correct airports and specific flight times, and the resulting compact Vega dataset did not contain any weather variables. Rather than carrying out an ineffective and probably wrong merge within the deadline of the project, it was decided to design the analysis based on the variables that actually exist.

However, the challenge of applying machine learning was kept. Specifically, the final analysis implements the engineering of a binary 30+ minute delay target, encoding of categorical flight features, a stratified train-test split and training of a balanced random-forest classifier. In terms of implementing the modeling part, simplifications were purposefully introduced in the modeling code. There was a previous version with multiple comparisons of models and more sophisticated preprocessing and cross-validation tools, but the current version relies only on `pandas.get_dummies`, `train_test_split` and one random forest for simplicity of the code explanation.

In addition, there was an increase in the scope of the data from the 77-row extract during EDA to 10,000 records. This change made airport, route, time-of-day, and model comparisons much more meaningful, while still keeping the project small enough to run quickly and reproduce in JupyterHub.

10. Work Plan Evaluation

Underestimation of the time spent on data collection and compatibility issues as compared to data analysis was a problem from the very beginning of the project. For example, the first exploratory data analysis was carried out using a tiny prejoined extract, while the initial idea about adding weather data provided by NOAA required the matching of the station and airport, as well as time synchronization, which was not completed in time. Eventually, the datasets were changed in such a way that the analysis could be repeated with one downloading script.

When the final dataset was chosen, the process of carrying out the descriptive analysis and visualization proceeded almost as expected. The most challenging part turned out to be the prediction model. In the first draft, more complex pipelines of scikit-learn and several models were used, which was simplified later due to the decrease in the runtime. Also, testing found some real errors caused by the last-minute editing: for example, a problem with the bin size when creating a miniature test data set, and an error in indenting of `analysis.py`. Fixing these before submission demonstrated the value of rerunning both direct tests and pytest after every significant change.

11. Testing

Tests were done to confirm the correctness of both real data set and small cases with known answers. The test program `test_project.py` first verifies that the final data set has exactly 10,000 records, and that the engineered features `delay_over_30`, `time_of_day`, and `day_of_week` columns have correct types. The missing value counts are compared directly to the results of pandas functions.

In addition, a small CSV of three rows is generated within a temporary directory. Since the delays, departure times, airports, and distances are selected manually, the expected results of the feature engineering are known beforehand. The tests confirm the 30 minute threshold, extracted hour, time of day bins, airport categorization, distance categorization, day of week summary, and route summary. The test CSV file is necessary because an algorithm may generate correct looking results for a large data set despite being wrong in some grouping or bin definitions.

The test program can be executed by using `python test_project.py` and pytest command. In the final project environment, pytest reports six successful tests. During the development process, code was constantly checked with flake8. Formatting errors, found during the final editing, were fixed before generating the archive. The combination of tests of small deterministic cases, tests of full data set, reproducible scripts, and CSV summary files is enough to confirm that the values of the report are generated by the code and not calculated manually.

12. Citations and Collaboration

Vega Datasets. `flights_10k` data package entry. https://vega.github.io/vega-datasets/datapackage.html

U.S. Bureau of Transportation Statistics. Reporting Carrier On-Time Performance database. https://www.transtats.bts.gov/

pandas documentation. https://pandas.pydata.org/docs/

Matplotlib documentation. https://matplotlib.org/stable/

scikit-learn documentation for RandomForestClassifier, train_test_split, and classification metrics. https://scikit-learn.org/stable/

CSE 163 Summer 2026 project/report requirements and code-quality guidance, University of Washington.

No other people were consulted outside course staff and normal course resources.